

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 2

Year: 2013

Subject: 394172 Mathematics II

Section: 15 - 16

Date: 25 December 2013

Time: 13.00- 15.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester.
Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Calculators are allowed in the examination.
4. This exam is a closed book examination.
5. Electronic communication devices are not allowed in the examination room.
6. The examination has 7 pages (including this page), 10 questions and a total score of 100 points.
7. Write all your solutions and answers on this examination sheet.

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Question 1. Make the *truth table* for $[(p \rightarrow (q \wedge r)) \rightarrow [r \leftrightarrow (p \vee q)]]$. The table with the different possible truth values has been provided. (10 points)

p	q	r
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

Question 2. Let p , q and r be statements. The compound statement $p \rightarrow (q \wedge r)$ is false and $p \rightarrow (q \wedge r)$ is true. Find the truth values of the following statement

$$[p \leftrightarrow (q \leftrightarrow r)] \wedge [(p \rightarrow r) \vee q]. \quad (10 \text{ points})$$

Question 3. Write and check *tautology* of the logical statement. (10 points)

$$(p \rightarrow r) \wedge (q \rightarrow r) \rightarrow [(p \vee q) \rightarrow r]$$

Question 4. Verify and check *equivalent* of the logical statements. (10 points)

$$(p \rightarrow r) \wedge (q \rightarrow r) \text{ and } (p \wedge q) \rightarrow r$$

Question 5. Given $U = \mathbb{R}$ (real numbers). Find the truth value of

$$\forall x [x^2 - 6x + 9 \geq 0] \Leftrightarrow \exists x [(x+5)(x-5) = x^2 - 25] \quad (10 \text{ points})$$

Question 6. Find the truth value of $\exists x \forall y [y^2x - 2xy + x = (y-1)^2]$
where $U = \{0, 1, 4, 9, 16, \dots\}$. (10 points)

Question 7. Convert the following numbers to a number with base of ten. (10 points)

7.1 $AEC_{16} =$

7.2 $(0.143)_5 =$

Question 8. Convert the following numbers to a number with a given base. (10 points)

8.1 $2135_7 =$ (base 12)

8.2 $12.312_4 =$ (base 16)

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Question 9. Simplify the following expressions. (10 points)

9.1 $(4213_5 - 412_5) \div 12_5$

9.2 $(F12AB_{16} - 3EABC_{16}) + FF6CDA_{16}$

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Question 10. Find the value of the expression with a given base. (10 points)

10.1 $(2321032_4 + 40275_8) - 2BA_{16}$ (base 8)

10.2 Solve the equation $(320)_4 = (110)_x$

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